

From individual alignment to institutional design

Aligned individual agents do not, by themselves, make a cooperative society

A. Familiar commons

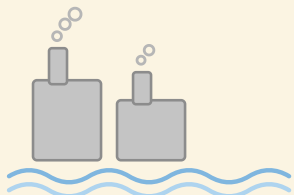
human social dilemmas



Shepherds

share a pasture

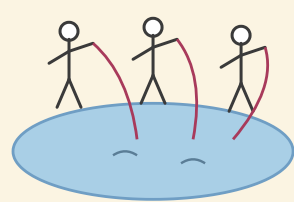
*overgrazing
depletes the field*



Factories

share a river

*pollution shifts
cost downstream*



Fisherman

share a pond

*overfishing cause
population collapse*

B. Artificial commons

in LLM-agent systems



Agents

share compute



power consumption

*coordinated load shifts
destabilize the cluster*



Agents

share databases

memory store

*unverified writes
corrupt the state*



Agents

share codebases



*unverified merges
break reliant applications*

C. Design space

institutional functions × mechanisms

Institutional functions

what an institution must do

reputation

monitoring

norms

repair

sanctions

constraints

adjudication

Mechanism families

how it shapes behavior

Game Mechanisms

constraint

epistemic

incentive

Institutional Mechanisms

normative

social

restorative

POSITION

Multi-agent AI safety requires **institutional design**. Even individually aligned agents can produce *collectively unsafe* outcomes when sharing scarce resources. Agent systems should therefore be evaluated as **governed systems** whose institutions shape collective behavior.

— our central claim

Contribution 1

Aligned individuals are not enough; institutions are needed.

Contribution 2

Distinguishing 7 functions from 6 mechanism families.

Contribution 3

An evaluation agenda for cooperation-shaping institutions.